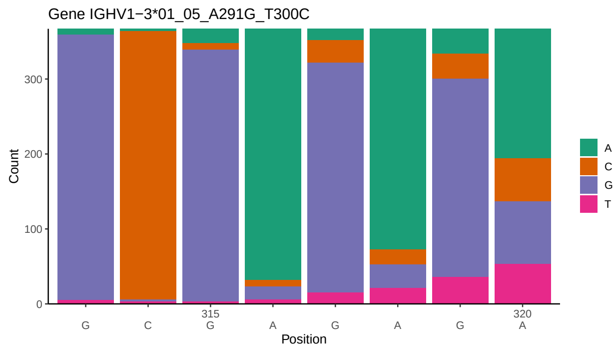
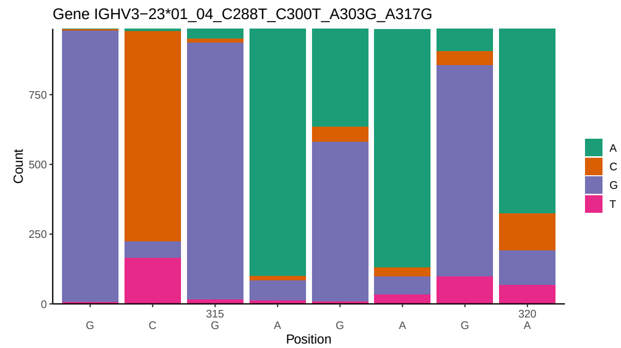
OGRDBstats Report

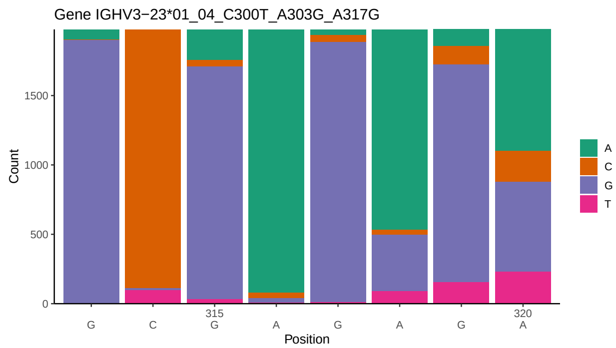
Contents

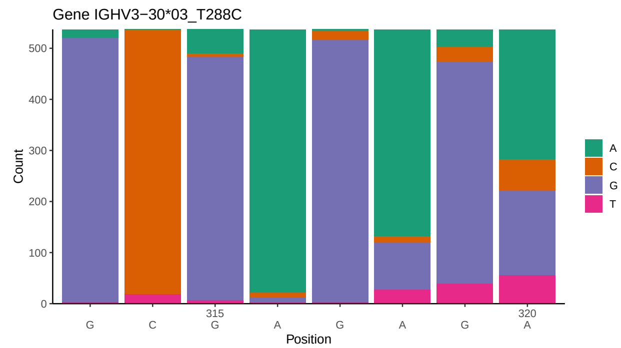
1 Novel sequence analysis

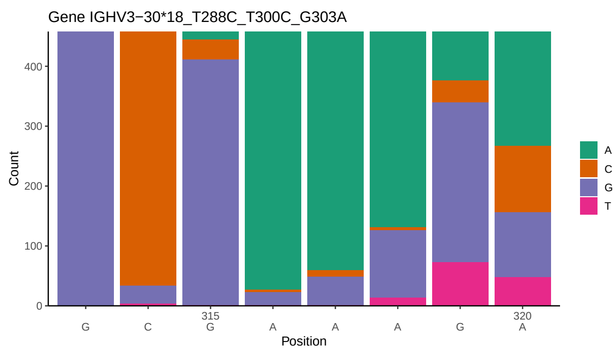
1.1 End-nucleotide composition

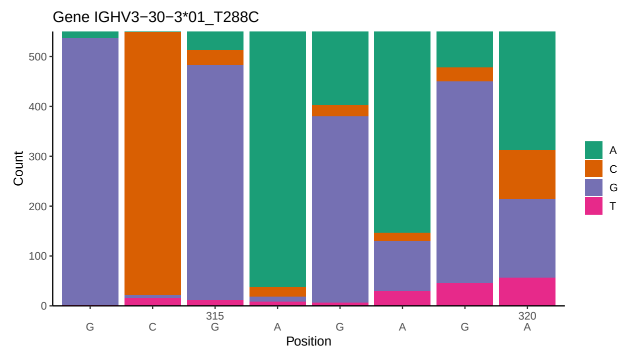


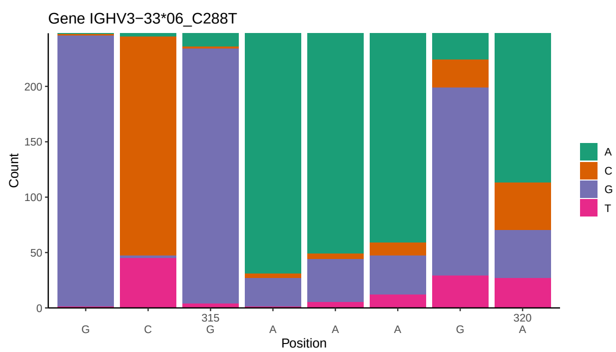


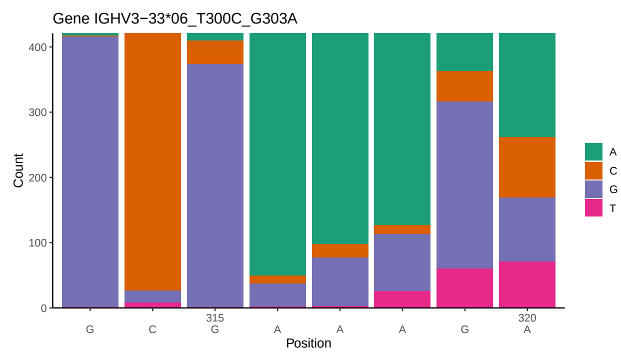


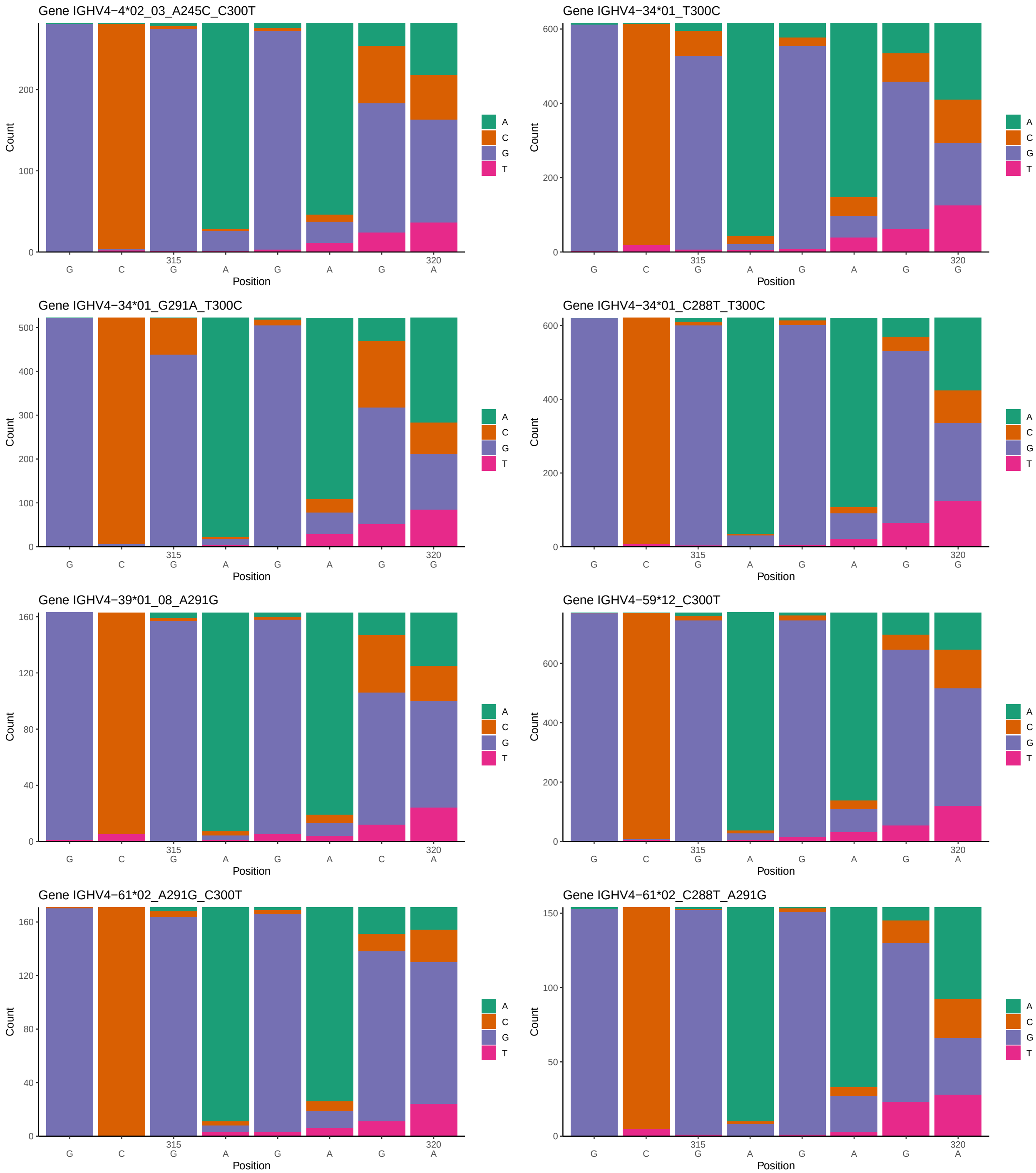


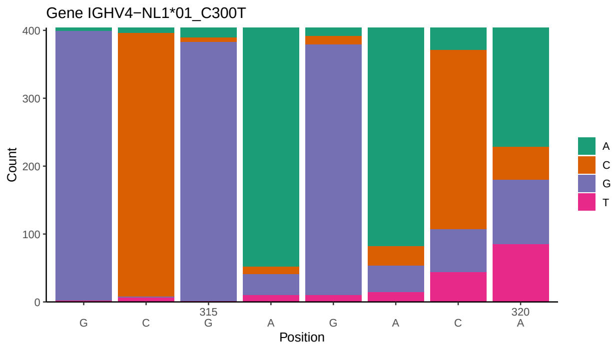




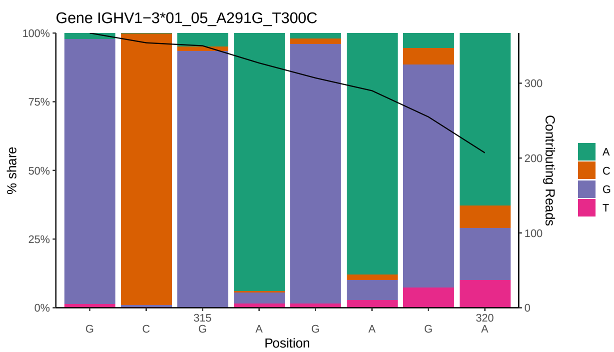


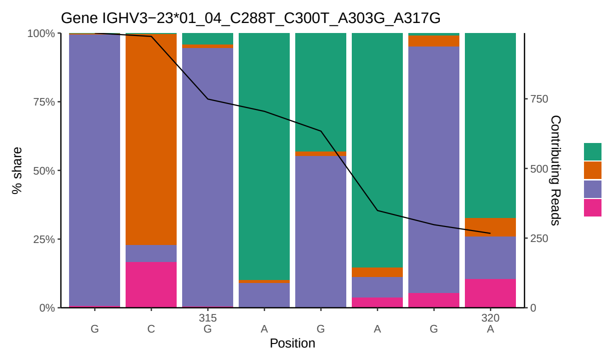


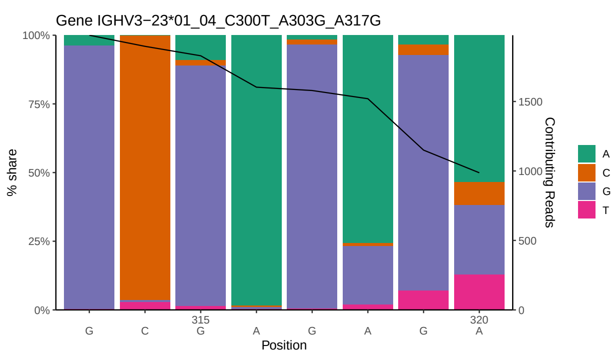

Gene IGHV4−4*02_03_A245C_C300T
Gene IGHV4−34*01_T300C
Gene IGHV4−34*01_G291A_T300C
Gene IGHV4−34*01_C288T_T300C
Gene IGHV4−39*01_08_A291G
Gene IGHV4−59*12_C300T
Gene IGHV4−61*02_A291G_C300T
Gene IGHV4−61*02_C288T_A291G
Count
Position
A
C
G
T
315
320

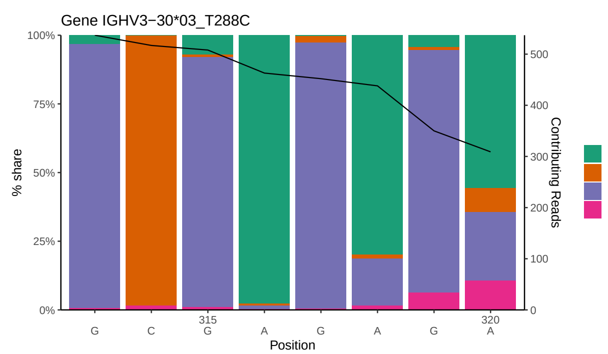


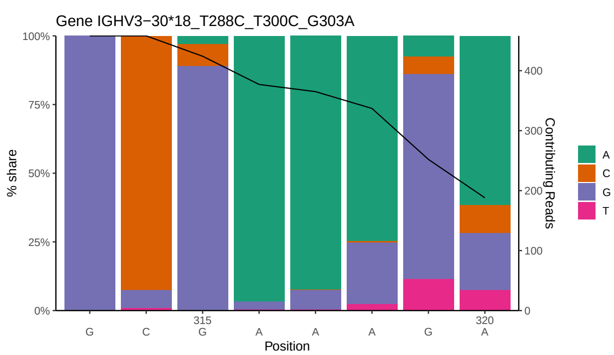
1.2 Per-nucleotide consensus where previous nucleotides match the consensus

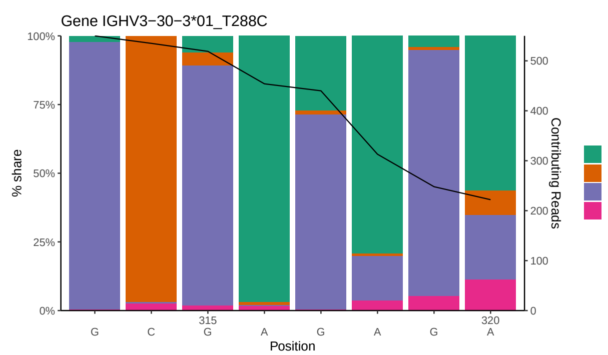


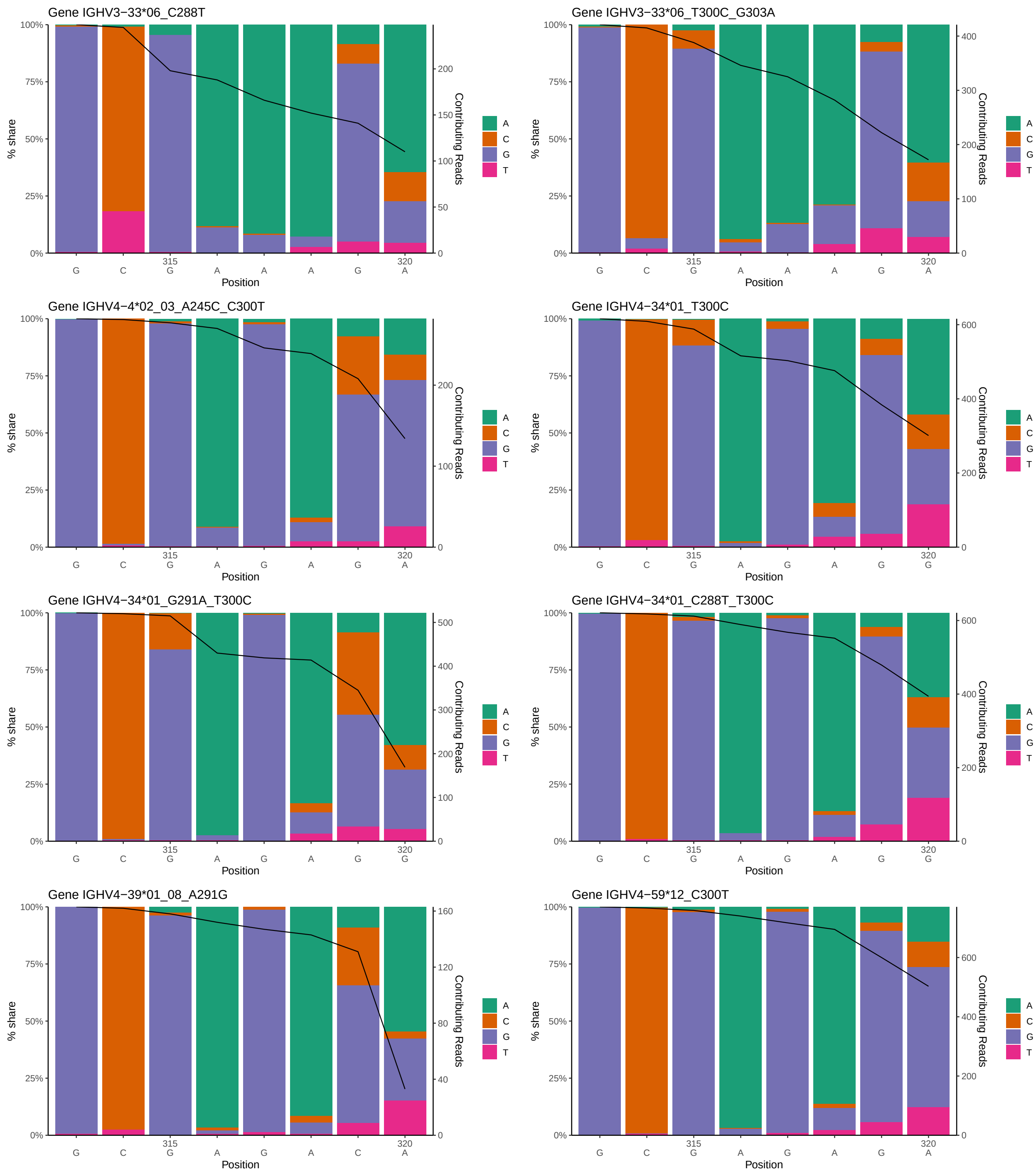




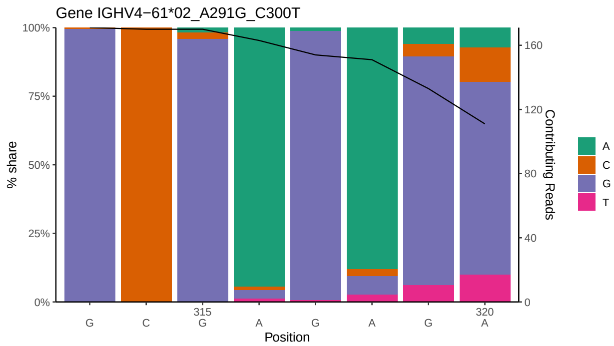


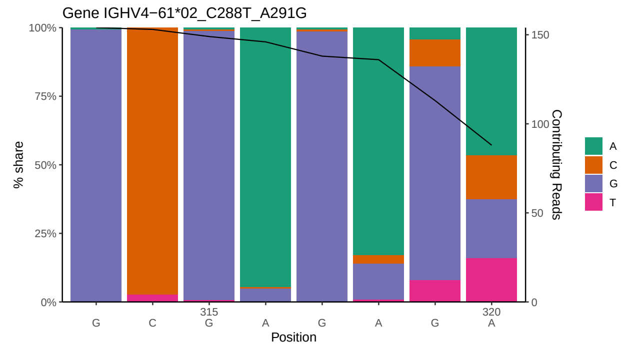


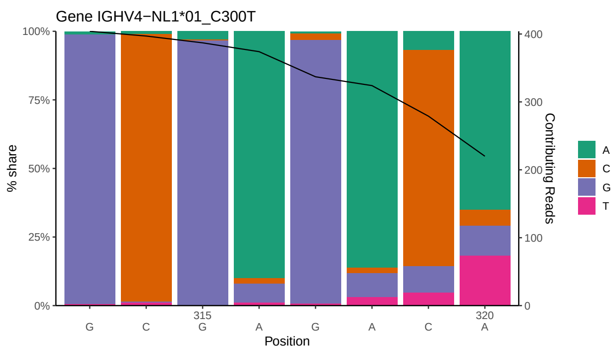




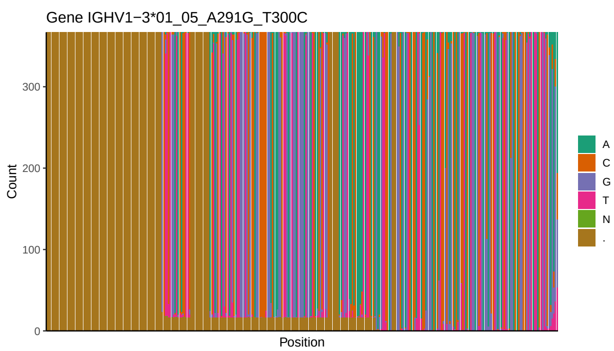
Gene IGHV3−33*06_C288T
Gene IGHV3−33*06_T300C_G303A
Gene IGHV4−4*02_03_A245C_C300T
Gene IGHV4−34*01_T300C
Gene IGHV4−34*01_G291A_T300C
Gene IGHV4−34*01_C288T_T300C
Gene IGHV4−39*01_08_A291G
Gene IGHV4−59*12_C300T
% share
Contributing Reads
Position
100%
75%
50%
25%
0%
315
320
A
C
G
T

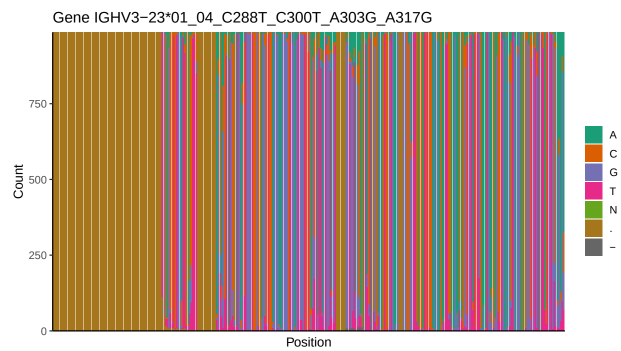


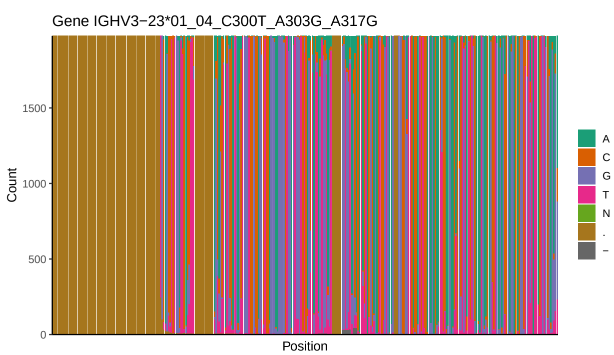


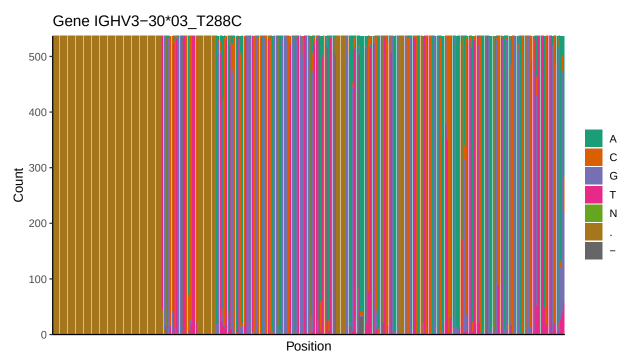


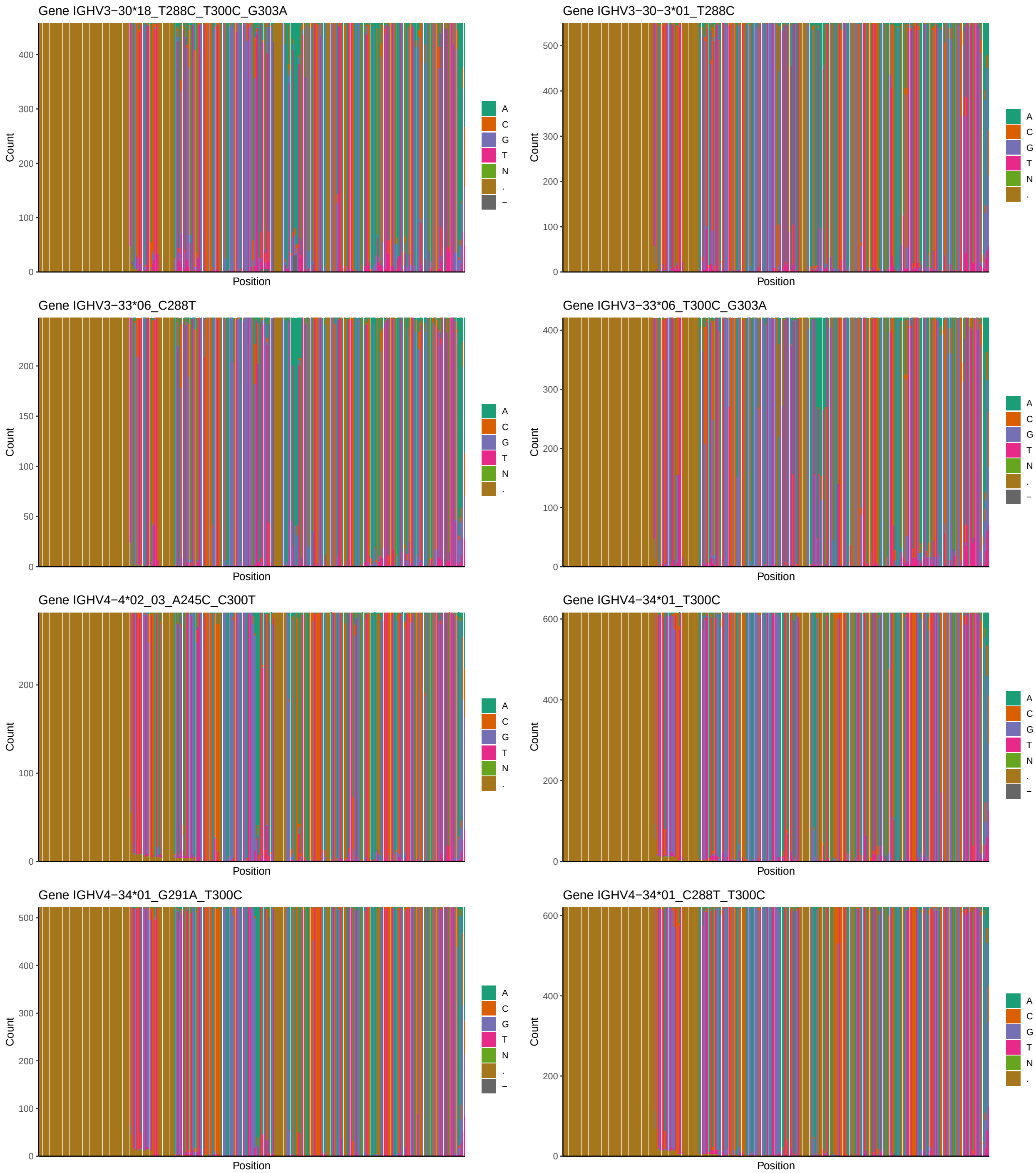
1.3 Whole-sequence composition of each assigned read

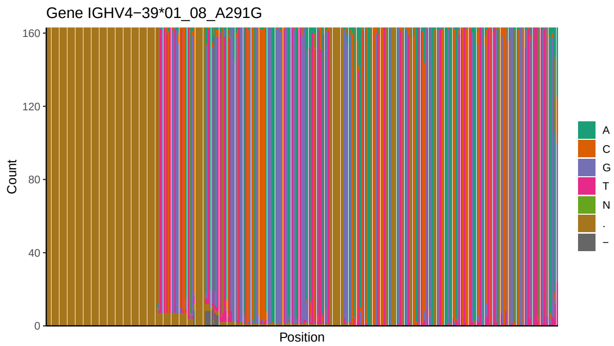


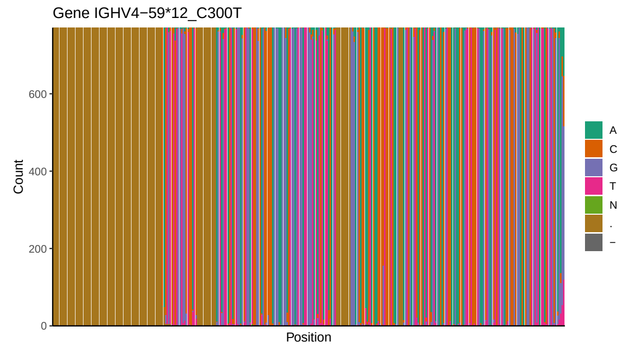


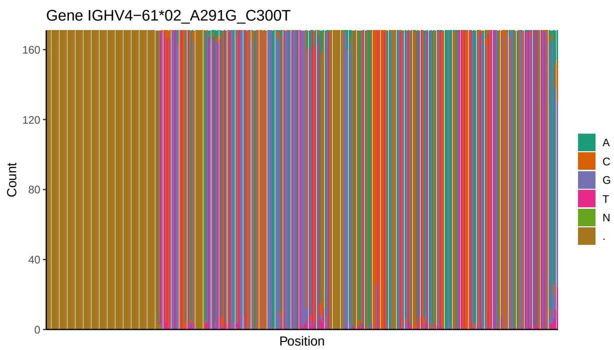


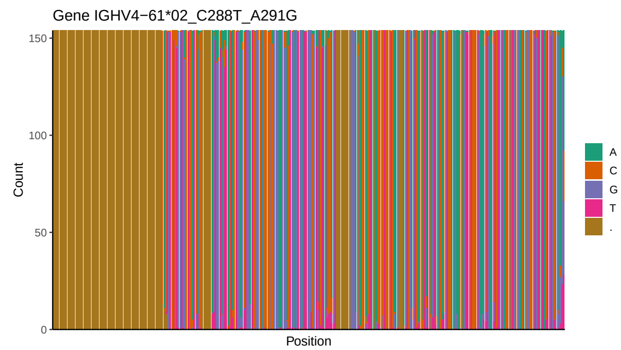


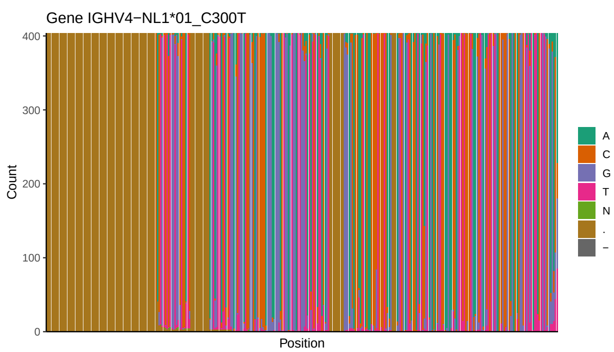

Gene IGHV3−30*18_T288C_T300C_G303A
Gene IGHV3−30−3*01_T288C
Gene IGHV3−33*06_C288T
Gene IGHV3−33*06_T300C_G303A
Gene IGHV4−4*02_03_A245C_C300T
Gene IGHV4−34*01_T300C
Gene IGHV4−34*01_G291A_T300C
Gene IGHV4−34*01_C288T_T300C
Count
Position
A
C
G
T
N
.
−



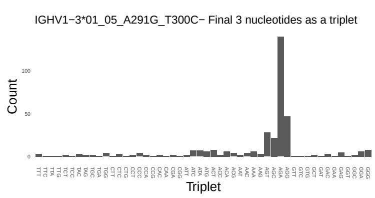


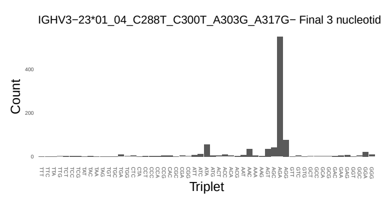


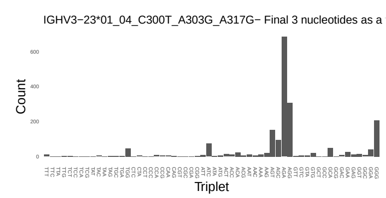


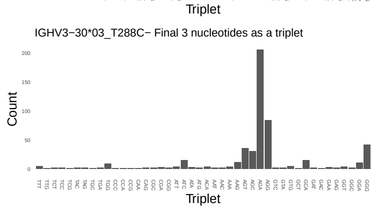


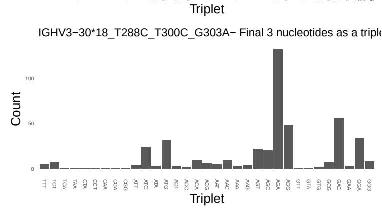
1.4 Final three nucleotides: frequency of each observed triplet

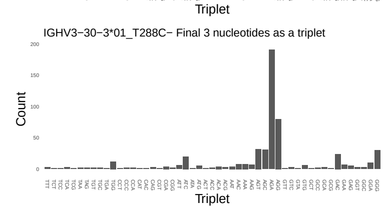


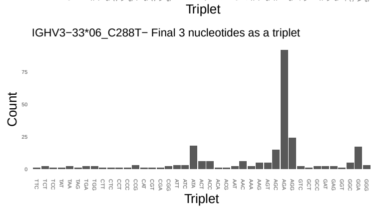


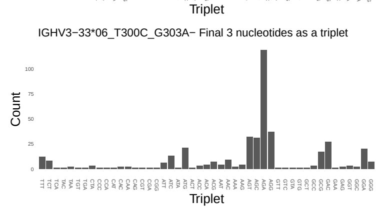


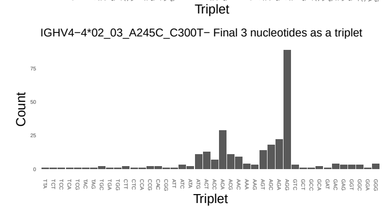


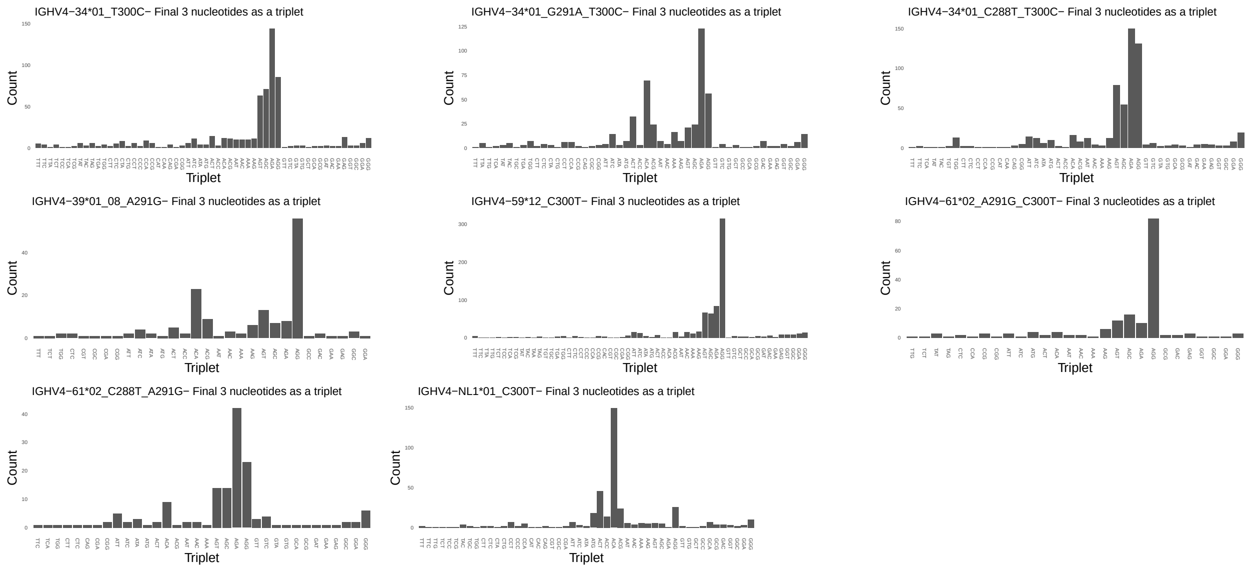




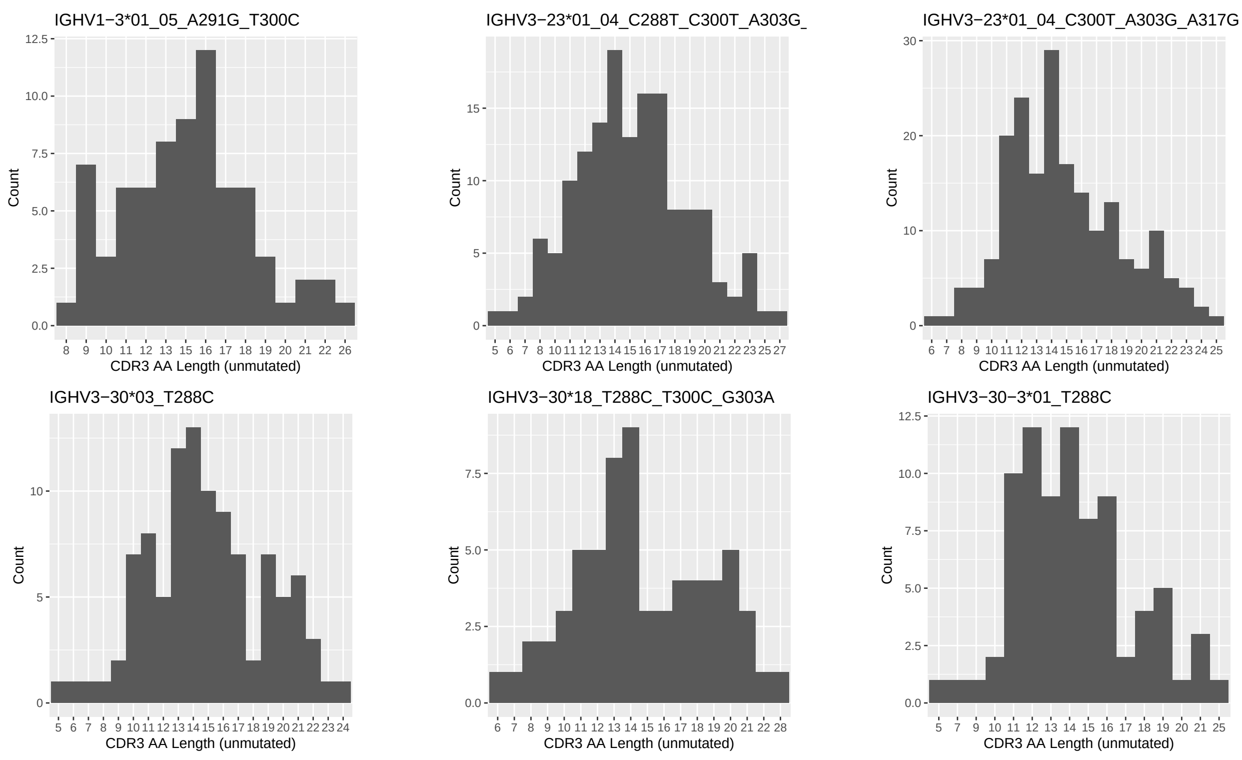


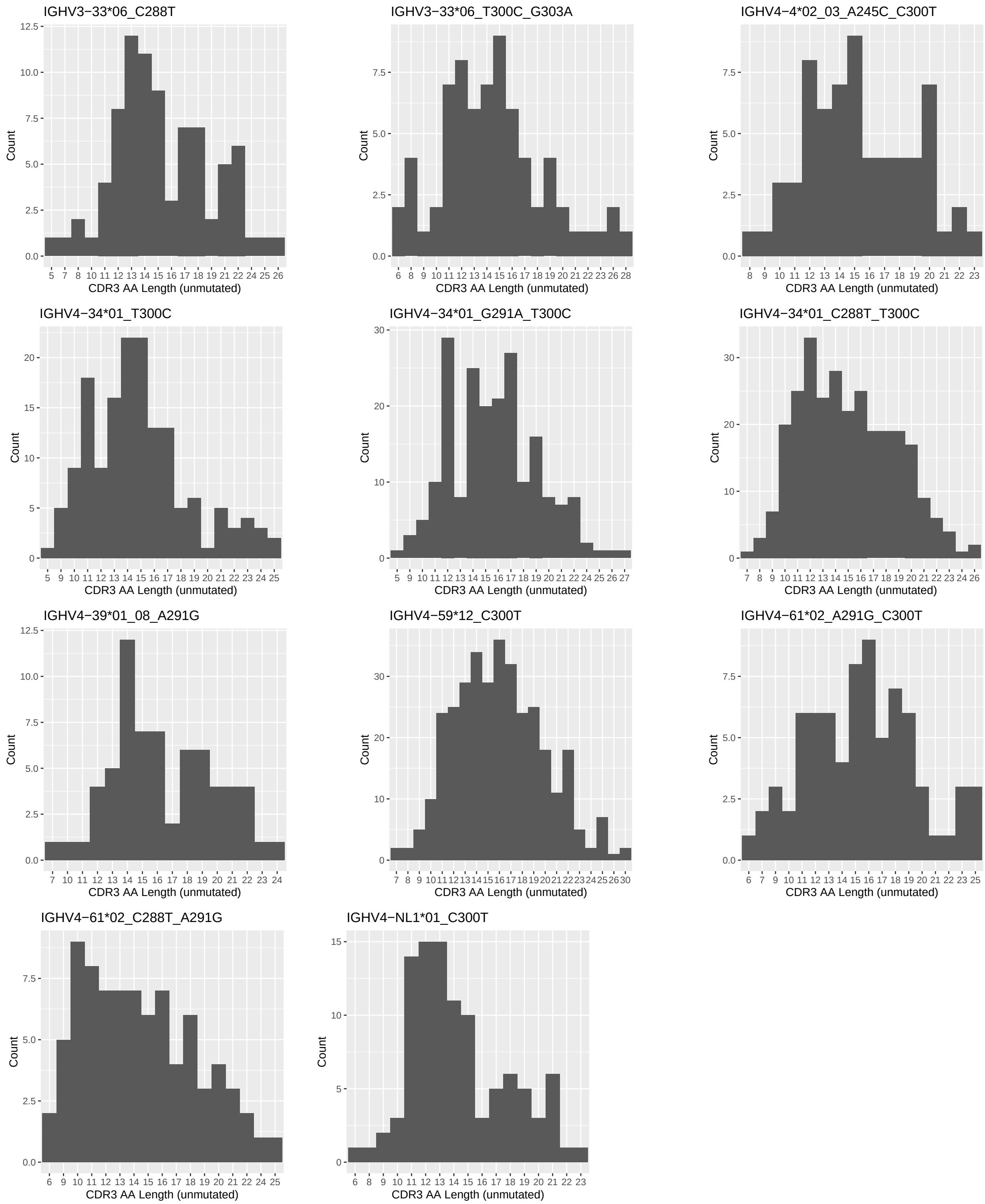




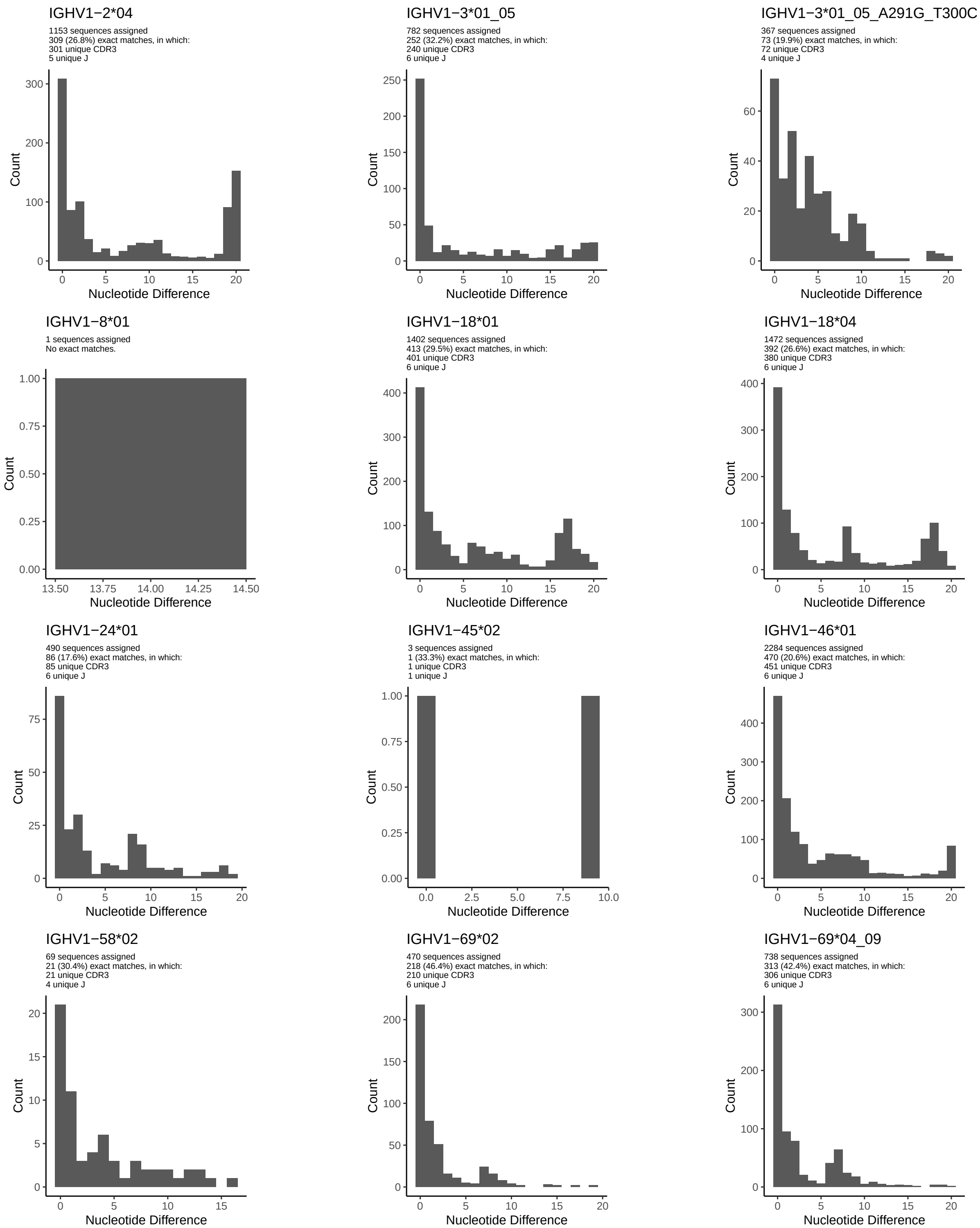


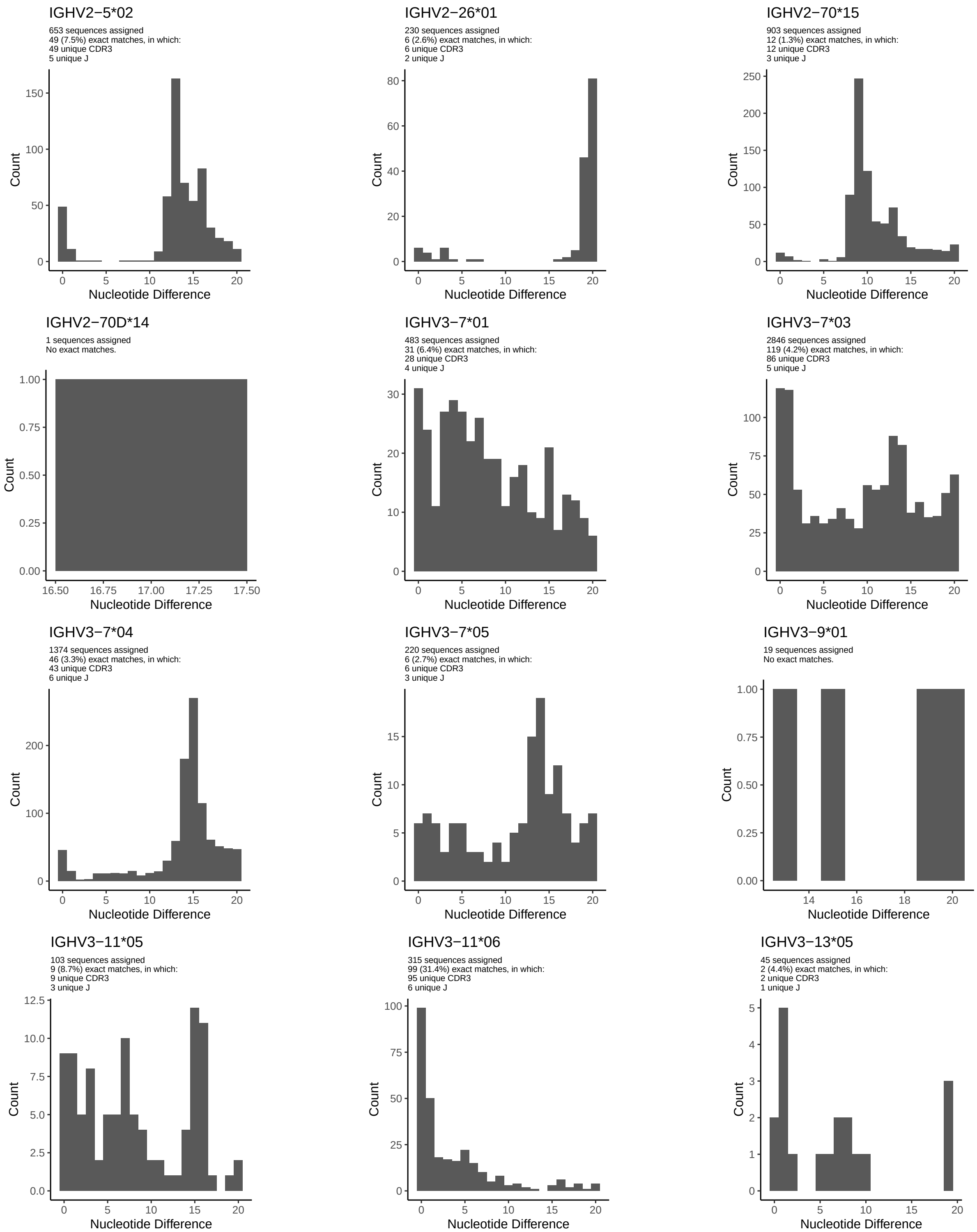
1.5 CDR3 length distribution, in assignments to novel alleles

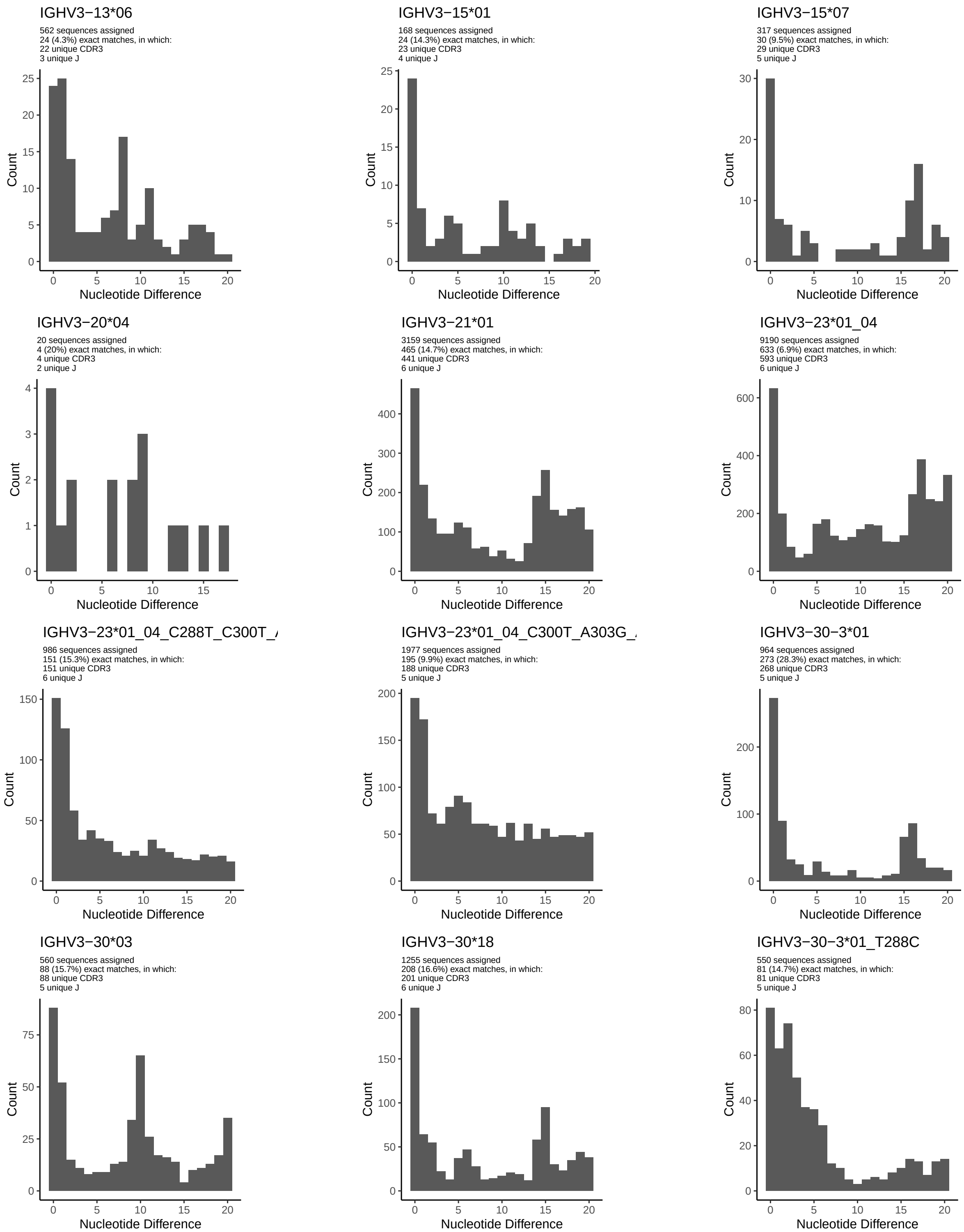


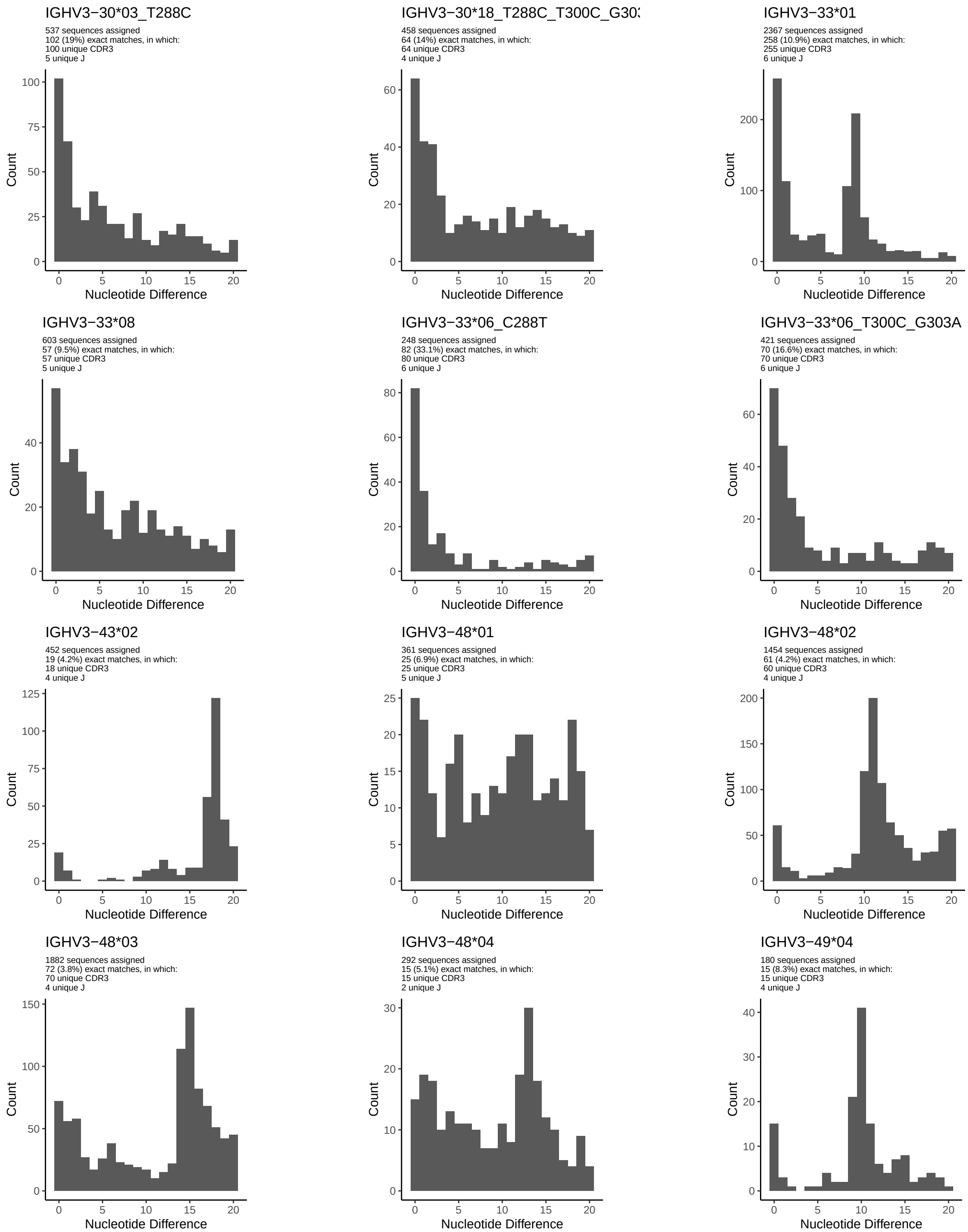

IGHV3−33*06_C288T
IGHV3−33*06_T300C_G303A
IGHV4−4*02_03_A245C_C300T
IGHV4−34*01_T300C
IGHV4−34*01_G291A_T300C
IGHV4−34*01_C288T_T300C
IGHV4−39*01_08_A291G
IGHV4−59*12_C300T
IGHV4−61*02_A291G_C300T
IGHV4−61*02_C288T_A291G
IGHV4−NL1*01_C300T
Count
CDR3 AA Length (unmutated)

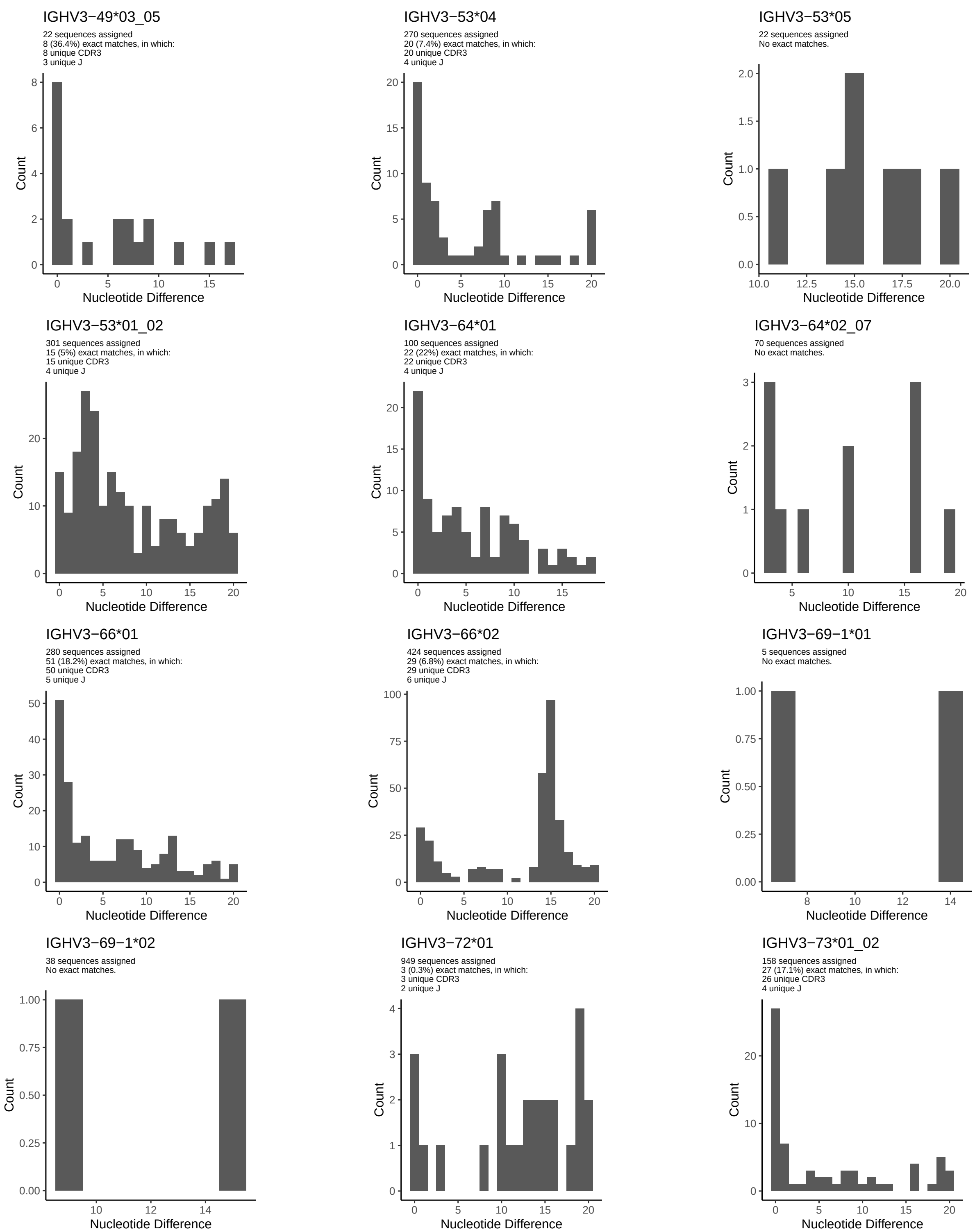
2 Variation from germline, in assignments to each allele

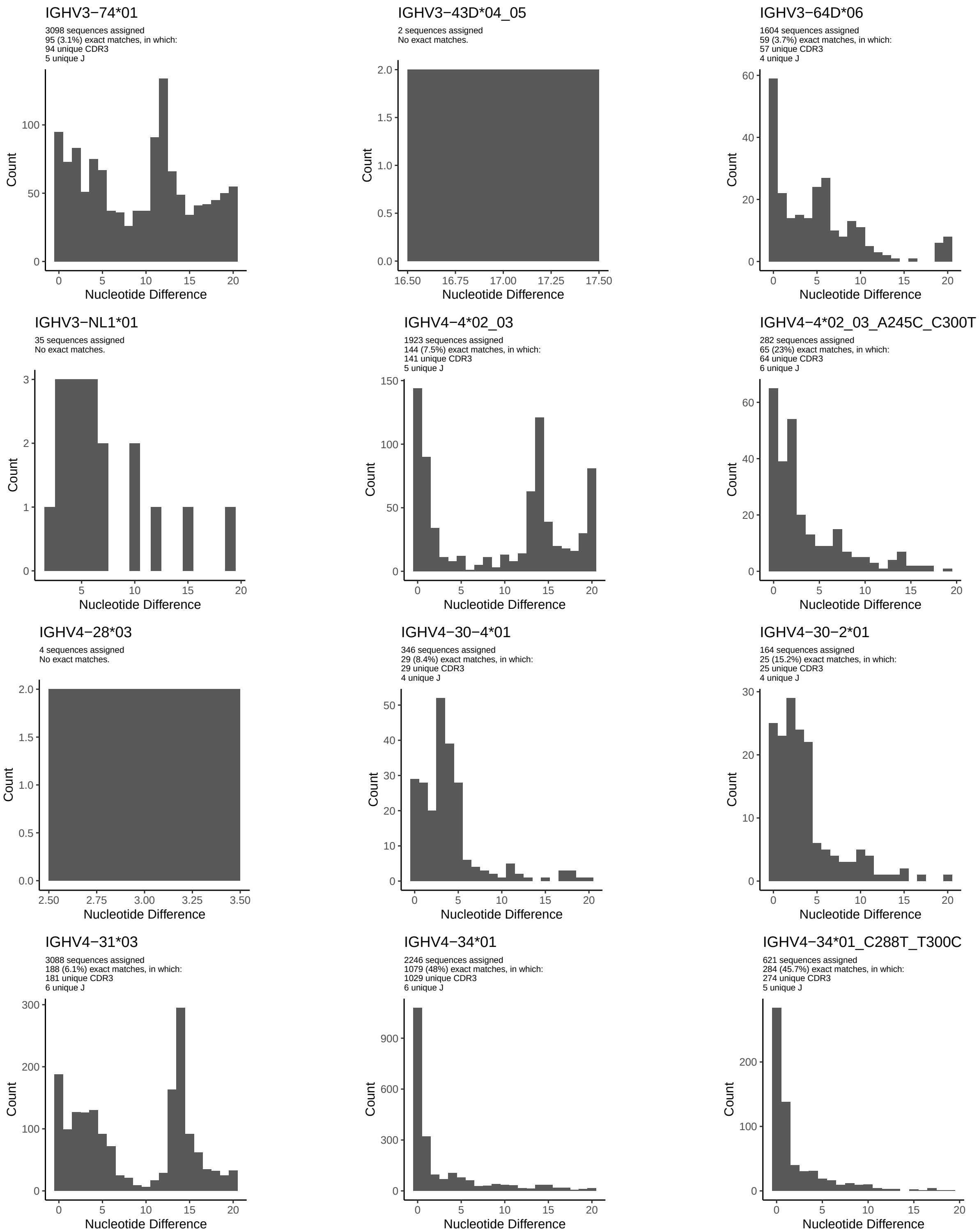


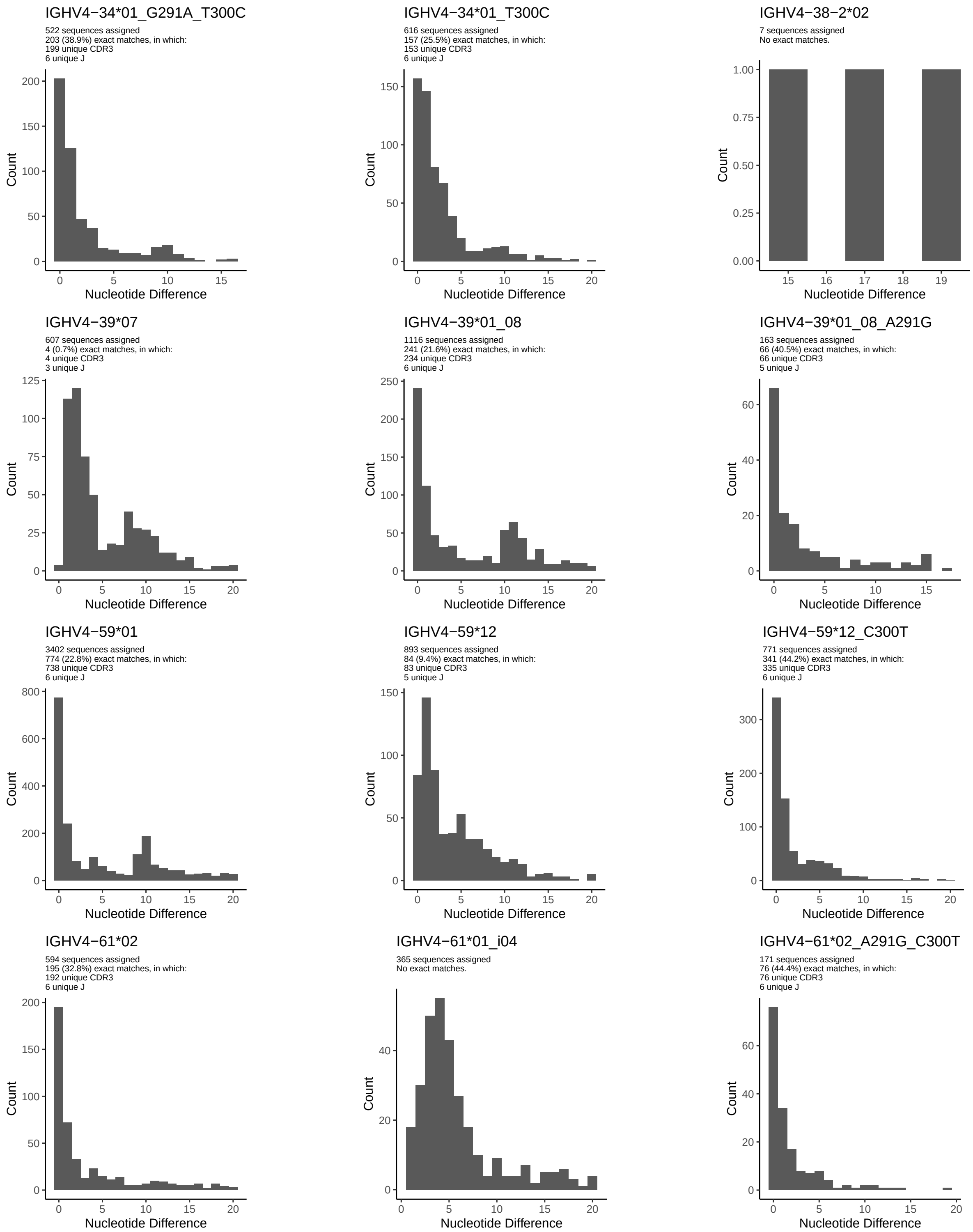

IGHV2−5*02
653 sequences assigned
49 (7.5%) exact matches, in which:
49 unique CDR3
5 unique J
IGHV2−26*01
230 sequences assigned
6 (2.6%) exact matches, in which:
6 unique CDR3
2 unique J
IGHV2−70*15
903 sequences assigned
12 (1.3%) exact matches, in which:
12 unique CDR3
3 unique J
IGHV2−70D*14
1 sequences assigned
No exact matches.
IGHV3−7*01
483 sequences assigned
31 (6.4%) exact matches, in which:
28 unique CDR3
4 unique J
IGHV3−7*03
2846 sequences assigned
119 (4.2%) exact matches, in which:
86 unique CDR3
5 unique J
IGHV3−7*04
1374 sequences assigned
46 (3.3%) exact matches, in which:
43 unique CDR3
6 unique J
IGHV3−7*05
220 sequences assigned
6 (2.7%) exact matches, in which:
6 unique CDR3
3 unique J
IGHV3−9*01
19 sequences assigned
No exact matches.
IGHV3−11*05
103 sequences assigned
9 (8.7%) exact matches, in which:
9 unique CDR3
3 unique J
IGHV3−11*06
315 sequences assigned
99 (31.4%) exact matches, in which:
95 unique CDR3
6 unique J
IGHV3−13*05
45 sequences assigned
2 (4.4%) exact matches, in which:
2 unique CDR3
1 unique J
Count
Nucleotide Difference

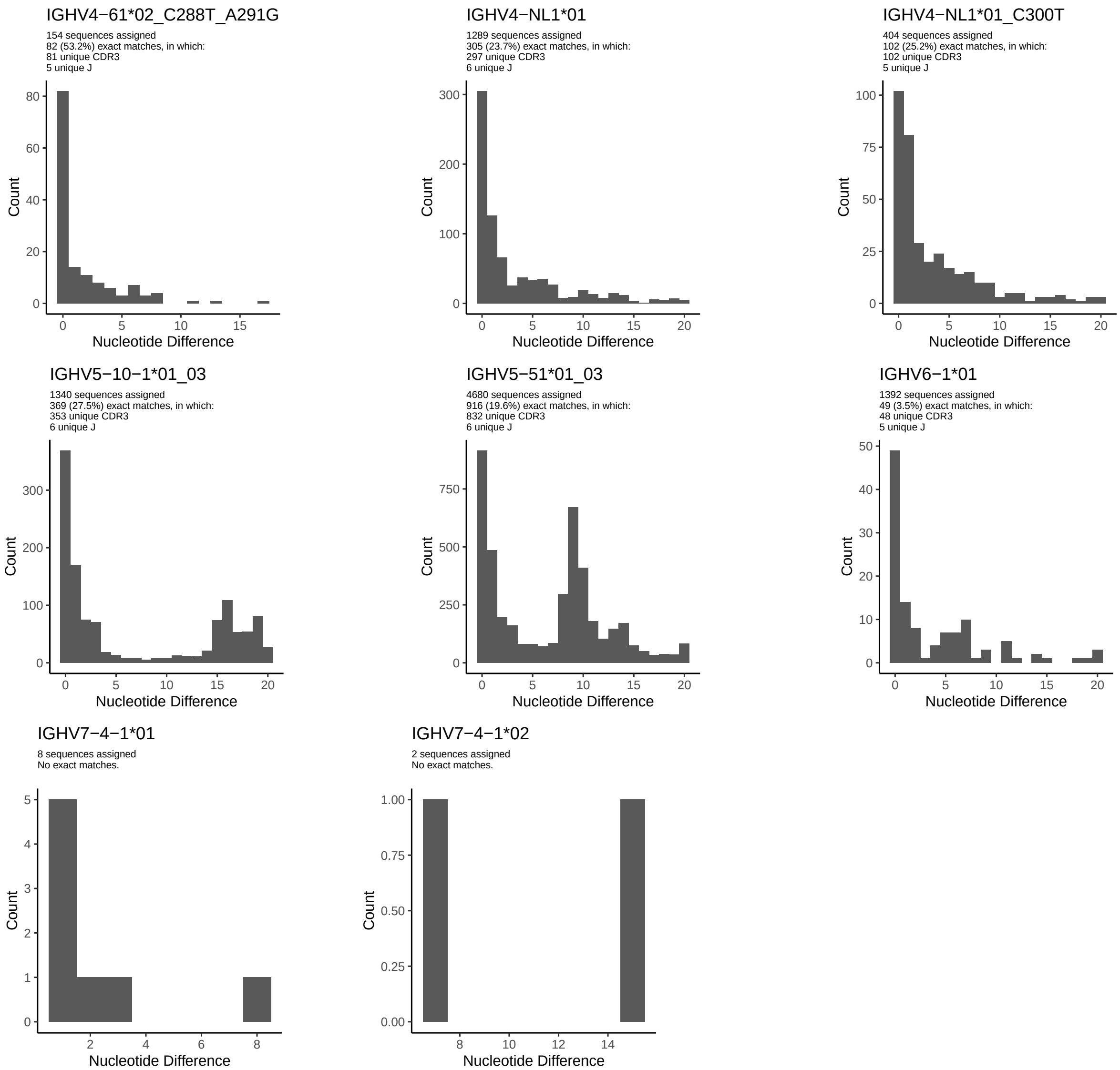

IGHV3−13*06
562 sequences assigned
24 (4.3%) exact matches, in which:
22 unique CDR3
3 unique J
IGHV3−15*01
168 sequences assigned
24 (14.3%) exact matches, in which:
23 unique CDR3
4 unique J
IGHV3−15*07
317 sequences assigned
30 (9.5%) exact matches, in which:
29 unique CDR3
5 unique J
IGHV3−20*04
20 sequences assigned
4 (20%) exact matches, in which:
4 unique CDR3
2 unique J
IGHV3−21*01
3159 sequences assigned
465 (14.7%) exact matches, in which:
441 unique CDR3
6 unique J
IGHV3−23*01_04
9190 sequences assigned
633 (6.9%) exact matches, in which:
593 unique CDR3
6 unique J
IGHV3−23*01_04_C288T_C300T_
986 sequences assigned
151 (15.3%) exact matches, in which:
151 unique CDR3
6 unique J
IGHV3−23*01_04_C300T_A303G_
1977 sequences assigned
195 (9.9%) exact matches, in which:
188 unique CDR3
5 unique J
IGHV3−30−3*01
964 sequences assigned
273 (28.3%) exact matches, in which:
268 unique CDR3
5 unique J
IGHV3−30*03
560 sequences assigned
88 (15.7%) exact matches, in which:
88 unique CDR3
5 unique J
IGHV3−30*18
1255 sequences assigned
208 (16.6%) exact matches, in which:
201 unique CDR3
6 unique J
IGHV3−30−3*01_T288C
550 sequences assigned
81 (14.7%) exact matches, in which:
81 unique CDR3
5 unique J
Count
Nucleotide Difference


IGHV3−30*03_T288C
537 sequences assigned
102 (19%) exact matches, in which:
100 unique CDR3
5 unique J
IGHV3−30*18_T288C_T300C_G30
458 sequences assigned
64 (14%) exact matches, in which:
64 unique CDR3
4 unique J
IGHV3−33*01
2367 sequences assigned
258 (10.9%) exact matches, in which:
255 unique CDR3
6 unique J
IGHV3−33*08
603 sequences assigned
57 (9.5%) exact matches, in which:
57 unique CDR3
5 unique J
IGHV3−33*06_C288T
248 sequences assigned
82 (33.1%) exact matches, in which:
80 unique CDR3
6 unique J
IGHV3−33*06_T300C_G303A
421 sequences assigned
70 (16.6%) exact matches, in which:
70 unique CDR3
6 unique J
IGHV3−43*02
452 sequences assigned
19 (4.2%) exact matches, in which:
18 unique CDR3
4 unique J
IGHV3−48*01
361 sequences assigned
25 (6.9%) exact matches, in which:
25 unique CDR3
5 unique J
IGHV3−48*02
1454 sequences assigned
61 (4.2%) exact matches, in which:
60 unique CDR3
4 unique J
IGHV3−48*03
1882 sequences assigned
72 (3.8%) exact matches, in which:
70 unique CDR3
4 unique J
IGHV3−48*04
292 sequences assigned
15 (5.1%) exact matches, in which:
15 unique CDR3
2 unique J
IGHV3−49*04
180 sequences assigned
15 (8.3%) exact matches, in which:
15 unique CDR3
4 unique J
Count
Nucleotide Difference

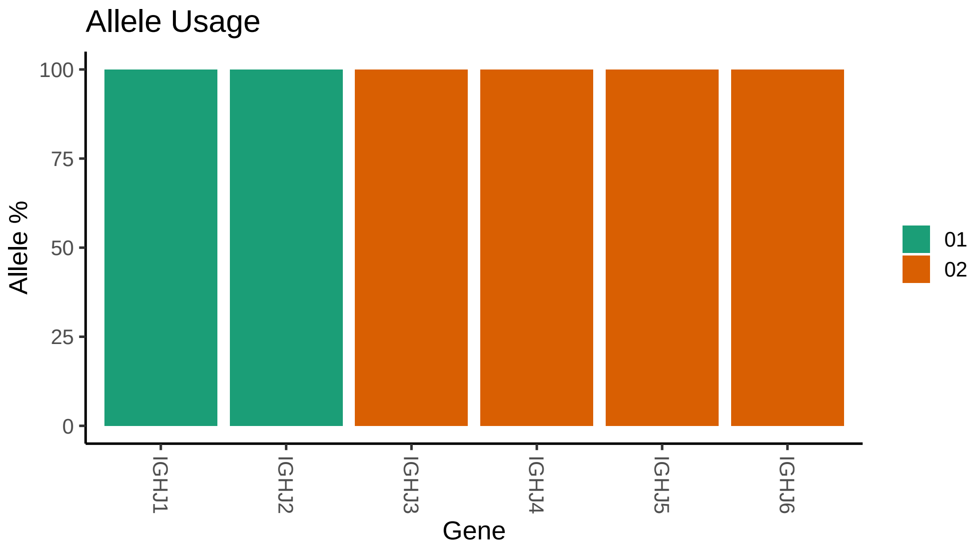

IGHV3−49*03_05
22 sequences assigned
8 (36.4%) exact matches, in which:
8 unique CDR3
3 unique J
IGHV3−53*04
270 sequences assigned
20 (7.4%) exact matches, in which:
20 unique CDR3
4 unique J
IGHV3−53*05
22 sequences assigned
No exact matches.
IGHV3−53*01_02
301 sequences assigned
15 (5%) exact matches, in which:
15 unique CDR3
4 unique J
IGHV3−64*01
100 sequences assigned
22 (22%) exact matches, in which:
22 unique CDR3
4 unique J
IGHV3−64*02_07
70 sequences assigned
No exact matches.
IGHV3−66*01
280 sequences assigned
51 (18.2%) exact matches, in which:
50 unique CDR3
5 unique J
IGHV3−66*02
424 sequences assigned
29 (6.8%) exact matches, in which:
29 unique CDR3
6 unique J
IGHV3−69−1*01
5 sequences assigned
No exact matches.
IGHV3−69−1*02
38 sequences assigned
No exact matches.
IGHV3−72*01
949 sequences assigned
3 (0.3%) exact matches, in which:
3 unique CDR3
2 unique J
IGHV3−73*01_02
158 sequences assigned
27 (17.1%) exact matches, in which:
26 unique CDR3
4 unique J
Count
Nucleotide Difference


IGHV3−74*01
3098 sequences assigned
95 (3.1%) exact matches, in which:
94 unique CDR3
5 unique J
IGHV3−43D*04_05
2 sequences assigned
No exact matches.
IGHV3−64D*06
1604 sequences assigned
59 (3.7%) exact matches, in which:
57 unique CDR3
4 unique J
IGHV3−NL1*01
35 sequences assigned
No exact matches.
IGHV4−4*02_03
1923 sequences assigned
144 (7.5%) exact matches, in which:
141 unique CDR3
5 unique J
IGHV4−4*02_03_A245C_C300T
282 sequences assigned
65 (23%) exact matches, in which:
64 unique CDR3
6 unique J
IGHV4−28*03
4 sequences assigned
No exact matches.
IGHV4−30−4*01
346 sequences assigned
29 (8.4%) exact matches, in which:
29 unique CDR3
4 unique J
IGHV4−30−2*01
164 sequences assigned
25 (15.2%) exact matches, in which:
25 unique CDR3
4 unique J
IGHV4−31*03
3088 sequences assigned
188 (6.1%) exact matches, in which:
181 unique CDR3
6 unique J
IGHV4−34*01
2246 sequences assigned
1079 (48%) exact matches, in which:
1029 unique CDR3
6 unique J
IGHV4−34*01_C288T_T300C
621 sequences assigned
284 (45.7%) exact matches, in which:
274 unique CDR3
5 unique J
Count
Nucleotide Difference


IGHV4−34*01_G291A_T300C
522 sequences assigned
203 (38.9%) exact matches, in which:
199 unique CDR3
6 unique J
IGHV4−34*01_T300C
616 sequences assigned
157 (25.5%) exact matches, in which:
153 unique CDR3
6 unique J
IGHV4−38−2*02
7 sequences assigned
No exact matches.
IGHV4−39*07
607 sequences assigned
4 (0.7%) exact matches, in which:
4 unique CDR3
3 unique J
IGHV4−39*01_08
1116 sequences assigned
241 (21.6%) exact matches, in which:
234 unique CDR3
6 unique J
IGHV4−39*01_08_A291G
163 sequences assigned
66 (40.5%) exact matches, in which:
66 unique CDR3
5 unique J
IGHV4−59*01
3402 sequences assigned
774 (22.8%) exact matches, in which:
738 unique CDR3
6 unique J
IGHV4−59*12
893 sequences assigned
84 (9.4%) exact matches, in which:
83 unique CDR3
5 unique J
IGHV4−59*12_C300T
771 sequences assigned
341 (44.2%) exact matches, in which:
335 unique CDR3
6 unique J
IGHV4−61*02
594 sequences assigned
195 (32.8%) exact matches, in which:
192 unique CDR3
6 unique J
IGHV4−61*01_i04
365 sequences assigned
No exact matches.
IGHV4−61*02_A291G_C300T
171 sequences assigned
76 (44.4%) exact matches, in which:
76 unique CDR3
6 unique J
Count
Nucleotide Difference


IGHV4−61*02_C288T_A291G
154 sequences assigned
82 (53.2%) exact matches, in which:
81 unique CDR3
5 unique J
IGHV4−NL1*01
1289 sequences assigned
305 (23.7%) exact matches, in which:
297 unique CDR3
6 unique J
IGHV4−NL1*01_C300T
404 sequences assigned
102 (25.2%) exact matches, in which:
102 unique CDR3
5 unique J
IGHV5−10−1*01_03
1340 sequences assigned
369 (27.5%) exact matches, in which:
353 unique CDR3
6 unique J
IGHV5−51*01_03
4680 sequences assigned
916 (19.6%) exact matches, in which:
832 unique CDR3
6 unique J
IGHV6−1*01
1392 sequences assigned
49 (3.5%) exact matches, in which:
48 unique CDR3
5 unique J
IGHV7−4−1*01
8 sequences assigned
No exact matches.
IGHV7−4−1*02
2 sequences assigned
No exact matches.
Count
Nucleotide Difference

3 Allele usage in potential haplotype anchor genes



4 Haplotype plots

5 Configuration settings

```
## Repertoire file: /misc/work/jenkins/PRJNA491287/M64 031/M64 031/M64 031/M64 031/bf/fe384729ac0544f01
##
## Germline reference file: /misc/work/jenkins/PRJNA491287/M64 031/M64 031/M64 031/M64 031/bf/fe384729a
##
## Novel allele file: /misc/work/jenkins/PRJNA491287/M64 031/M64 031/M64 031/M64 031/bf/fe384729ac0544f
##
## Species: Homosapiens
##
## Chain: IGHV
##
## Segment: V
##
## Unexpected N in reference sequence IGHV1 3 01_05: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 3 01_05_A291G_T300C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 23 01_04: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 23 01_04_C288T_C300T_A303G_A317G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 23 01_04_C300T_A303G_A317G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 30 03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 03_T288C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 30 18: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 18_T288C_T300C_G303A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 30 3 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 3 01_T288C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 33 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 33 06_C288T: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 33 06_T300C_G303A: replacing with gap
```

```
##
##
## Unexpected N in reference sequence IGHV4 4 02_03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 4 02_03_A245C_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 34 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 34 01_T300C: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 34 01_G291A_T300C: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 34 01_C288T_T300C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 39 01_08: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 39 01_08_A291G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 59 12: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 59 12_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 61 02: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 61 02_A291G_C300T: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 61 02_C288T_A291G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 NL1 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 NL1 01_C300T: replacing with gap
```